

Project Proposal: Generative EEG Foundation Model for Channel Reconstruction

Problem Statement: Current EEG foundation models such as NeuroAtlas, EEG-FM Compass, and BrainFM, focus primarily on representation learning and cross-task generalization. However, these models assume access to full multi-channel EEG data, limiting their applicability in low-cost or portable systems where only sparse channels are available. Missing channels reduce spatial resolution and hinder accurate interpretation of neural dynamics. This project addresses the need for a generative approach that reconstructs missing EEG channels from limited input using learned spatial–temporal correlations. Our primary goal is to lower the cost of access to EEG devices.

Proposed Contribution: This work introduces a Spatial Generative EEG Foundation Model that learns inter-channel dependencies and reconstructs missing EEG signals from sparse recordings. The model extends self-supervised pre-training objectives with spatial masking and conditional generation, enabling channel-level imputation. By doing so, it supports low-cost EEG acquisition and enhances downstream performance in emotion recognition, workload estimation, and clinical monitoring.

Contrast with Existing Models

Table 1: Comparing the paradigm of our proposed with the paradigm of existing models

Dimension	Existing Work	Proposed Novelty
Objective	Representation learning and cross-task generalization	Spatial generative modeling for channel reconstruction
Data Utilization	Full multi-channel EEG datasets	Sparse EEG data with learned channel generation
Learning Paradigm	Masked signal reconstruction, contrastive learning	Spatial masking and conditional generation
Evaluation Metrics	Downstream task accuracy	Spatial fidelity, spectral coherence, and downstream improvement
Impact	Generalization and labeling efficiency	Low-cost EEG acquisition and hardware-efficient BCI systems

Table 2: Summary of Existing EEG Foundation Models

Model / Framework	Year	Lead Institution(s)	Core Capability / Focus	Key Features
NeuroAtlas ^[1]	2026	KU Leuven & MIT	Unified benchmarking across clinical and BCI tasks	42 datasets; evaluates epilepsy, sleep, brain aging, and BCI paradigms
EEG-FM Compass ^[2]	2026	Huazhong Univ. of Sci. & Tech.	Comprehensive taxonomy and benchmarking	Defines pre-training objectives; scaling law analysis
EEG-FM Benchmark ^[3]	2025 - 2026	Tongji University	Comparative evaluation of 12 open-source EEG FMs	Tests cross-subject generalization and fine-tuning strategies
Simple Review of EEG Foundation Models ^[4]	2025	Zhejiang University	Survey of datasets and architectures	Summarizes early EEG-FMs; identifies emotion and motor imagery applications
BrainFM ^[5]	2026	Various labs	Self-supervised EEG representation learning	Masked signal reconstruction; cross-paradigm transfer; supports emotion, sleep, motor imagery decoding

Expected Outcomes

The project will deliver a generative EEG foundation model capable of reconstructing missing channels from sparse data. This innovation will advance hardware-efficient brain–computer interfaces, improve signal fidelity, and expand access to EEG-based research and clinical applications.

Execution Plan

We first develop and evaluate a generative EEG foundation model for channel reconstruction on data from a single manufacturer. Next, we investigate adaptation and fine-tuning strategies to transfer this model across different EEG devices with distinct calibration and reference schemes. Finally, we extend the approach to a manufacturer-agnostic model that learns device-invariant neural dynamics while

preserving manufacturer-specific channel characteristics, enabling robust channel generation across heterogeneous hardware.

References

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